THE RIGHT ADJOINT OF THE BASE CHANGE FUNCTOR OVER A FIBRE PRODUCT OF RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The functor from modules over a fibre product of rings to pairs of modules appearing in the statement. the author refers to the displayed equation equation-modules, and the surrounding section, source lines 400-461. Supplies the functor whose right adjoint is constructed.
- The tensor-restriction adjunction identifying maps out of a base changed module with maps out of the original one. the author cites Algebra, Lemma algebra-lemma-adjoint-tensor-restrict at proof lines 495-497. Rewrites the right hand side of the adjunction equality.
- Fibre products of sets and of modules and elementwise verification of inverse constructions. used at proof lines 499-510. Establishes that the two constructions are mutually inverse.

Source reference. Original lemma 0D2F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16380–16380. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0D2E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16379–16379. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05DQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6963–6963. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Given a commutative diagram of rings*

$$\begin{array}{ccc} R & \longleftarrow & R' \\ \uparrow & & \uparrow \\ B & \longleftarrow & B' \end{array}$$

the functor (0D2E) has a right adjoint, namely the functor

$$F : (N, M', \varphi) \longmapsto N \times_\varphi M'$$

(see proof for elucidation).

Proof. Given an object (N, M', φ) of the category $\mathrm{Mod}_B \times_{\mathrm{Mod}_R} \mathrm{Mod}_{R'}$ we set

$$N \times_\varphi M' = \{(n, m') \in N \times M' \mid \varphi(n \otimes 1) = m' \otimes 1 \text{ in } M' \otimes_{R'} R\}$$

viewed as a B'-module. The adjointness statement is that for a B'-module L' and a triple (N, M', φ) we have

$$\mathrm{Hom}_{B'}(L', N \times_\varphi M') = \mathrm{Hom}_B(L' \otimes_{B'} B, N) \times_{\mathrm{Hom}_R(L' \otimes_{B'} R, M' \otimes_{R'} R)} \mathrm{Hom}_{R'}(L' \otimes_{B'} R', M')$$

By Algebra, Lemma 05DQ the right hand side is equal to

$$\mathrm{Hom}_{B'}(L', N) \times_{\mathrm{Hom}_{B'}(L', M' \otimes_{R'} R)} \mathrm{Hom}_{B'}(L', M')$$

Thus it is clear that for a pair (g, f') of elements of this fibre product we get an B'-linear map $L' \to N \times_\varphi M'$, $l' \mapsto (g(l'), f'(l'))$. Conversely, given a B' linear map $g' : L' \to N \times_\varphi M'$ we can set g equal to the composition $L' \to N \times_\varphi M' \to N$ and f' equal to the composition $L' \to N \times_\varphi M' \to M'$. These constructions are mutually inverse to each other and define the desired isomorphism. □